Alexis Christoforides

Boston Area, MA USA

[email](#) - [Github](#) - [Google Scholar](#) - [LinkedIn](#)

Expert software engineer offering more than 15 years of professional software development experience & enthusiasm for the craft. I've been a senior software engineer, solo developer or a project leader for scientific analysis software, programming language runtimes, API designs, embedded device firmware & more.

Strong experience & background in C, Python, C#, shell scripts & Java. Very comfortable using LLM coding assistance.

Career & Highlights

2020 - Present

Meadow.OS Team Engineering Lead at Wilderness Labs

I lead the effort developing the system software for the Meadow IoT single-board computer which lets developers use C# & .NET to rapidly write secure & performant applications for devices with limited resources & connectivity.

- Ported the Mono .NET runtime to run on an STM32F7 MCU, the NuttX RTOS, <32MB of RAM & <32MB of flash
- Integrated mbedTLS to Mono to serve as a new MCU-friendly TLS provider
- Designed & implemented full-system (firmware + app), assured & secure over-the-air (OTA) update machinery, spanning from custom bootloader to reliable network messaging logic using MQTT & written in C#
- Co-designed & co-developed our IoT app lifecycle for resilience, live monitoring, crash recovery & update/rollback support
- Co-designed our low-level I/O API for .NET (GPIO, SPI, I2C, interrupt processing)
- Ported TensorFlow Lite Micro to the Meadow & wrote bindings for .NET
- Maintained our CI jobs & Azure Pipelines-based OS release process
- Guided the team through developing OS features & resolving user issues

2014 - 2020

Mono Runtime Engineer at Xamarin & Microsoft

I was part of the **Mono** runtime engine team, an open-source & cross-platform .NET implementation used for C# & F#-based apps on iOS, Android, video game consoles, & other small devices. Xamarin was acquired by Microsoft in 2016 where my work continued.

- Improved the "SGen" generational garbage collector, e.g. by reducing memory allocation for the string type & adding "canary guard"-style diagnostics to the heap
- Improved the reliability & diagnostic capabilities of Mono's native crash report machinery
- Ported the file & network I/O subsystem from Microsoft's CoreFX library to Mono
- Maintained the CI & Release packaging software for the Mono MacOS SDK, a >1GB installable archive.
- Collaborated with the product teams to integrate the latest Mono development into the Xamarin Android/iOS products

2009 - 2014

Bioinformatician & Software Engineer at TGen

I wrote research & medical-use analysis software & methods for whole-genome sequencing data (using Java, C++, Python, shell scripts, R, etc.). Specialized in cancer genomics & human genome sequencing analysis.

- Developed a method for comparing DNA/RNA between tumor & normal samples in patients with solid cancers. Wrote & maintained the software implementation in Java & published a paper
- Was a small part of the 1000 Genomes pilot project, which created the most comprehensive variant map of the human genome at the time & created a public dataset used as reference in hundreds of subsequent studies (see Publications)
- Helped design, maintain, run & troubleshoot TGen's HPC cancer genomics analysis pipeline & genomic data store
- Created a custom web portal for collaborative version-tracked data entry & report generation of confidential cancer patient data
- Provided whole-genome data analysis & internal software support to PIs & studies throughout the institute

Higher Education

Computer Science BSc
at Arizona State University, Tempe AZ

Biomedical Informatics Ph. D. (withdrawn)
at Arizona State University/Mayo Clinic

(Completed all coursework & published one first-author paper before withdrawing)

Most Important Publications

Christoforides, A., Carpten, J. D., Weiss, G. J., Demeure, M. J., Hoff, D. D. V., & Craig, D. W. (2013). **Identification of somatic mutations in cancer through Bayesian-based analysis of sequenced genome pairs**. *BMC Genomics*, 14(1), 302. doi:10.1186/1471-2164-14-302

(2010). **A map of human genome variation from population-scale sequencing**. *Nature*, 467(7319), 1061–1073. doi:10.1038/nature09534